

## MIT Lab Assignment – 10/09/26

Maintaining the same assignment submission guidelines you already know, solve the following problems and upload it in the classroom by **today, 11:59 pm**. If you don't know about the topics learn first and then solve accordingly.

### P1: Orthogonal Projection onto a Subspace

Let  $V = \text{span}\{(1,1,0), (1,0,1)\} \subset \mathbb{R}^3$ . Find the **orthogonal projection** of the vector  $x=(1,2,3)$  onto  $V$ .

### P2: Orthogonal Complement

Let  $U = \text{span}\{(1,2,1), (0,1,-1)\}$  in  $\mathbb{R}^3$ . Find an orthonormal basis for  $U^\perp$ .

### P3: Orthogonal Matrix Check

Let  $Q$  is a matrix which is shown here, show that  $Q$  is orthogonal.

$$Q = \frac{1}{3} \begin{bmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{bmatrix}$$

### P4: Orthogonal Diagonalization

Let  $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ . Diagonalize  $A$  using an orthogonal matrix.

### P5: Distance to a Subspace

Let  $S = \text{span}\{(1,1,1)\}$ . Find the shortest distance from  $x=(2,1,3)$  to  $S$ .